



GTC4Lusso USA - Engine intake manifold - Latest Update: 2021-01-26

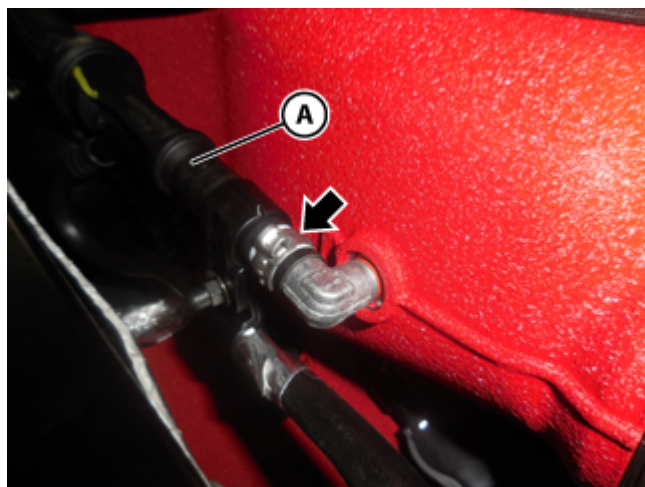
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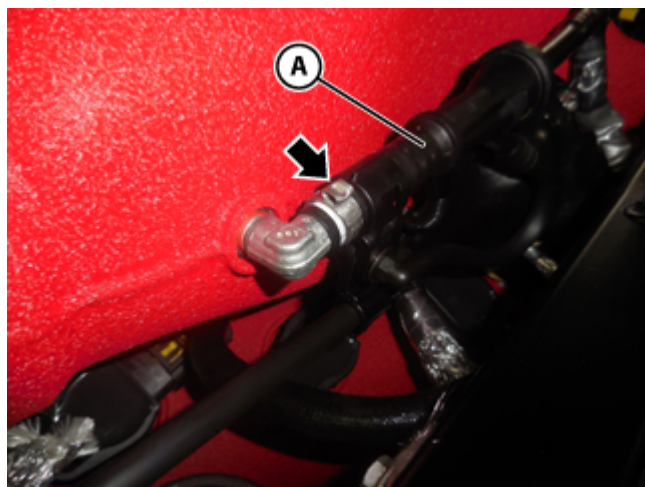
B4.04 Engine intake manifold

Removing the engine intake manifold

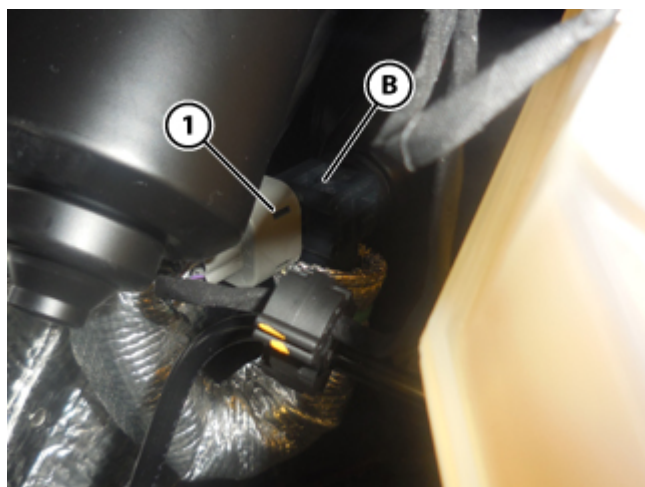
- Disconnect the battery ( F2.01).
- Remove the engine compartment cosmetic shields ( E3.14).
- Remove the lateral underwindscreen cosmetic shields.
- Remove the air flow meters ( F3.08).
- Remove the motorised throttle bodies ( F3.09).
- Remove the windscreen wiper unit ( E3.10).



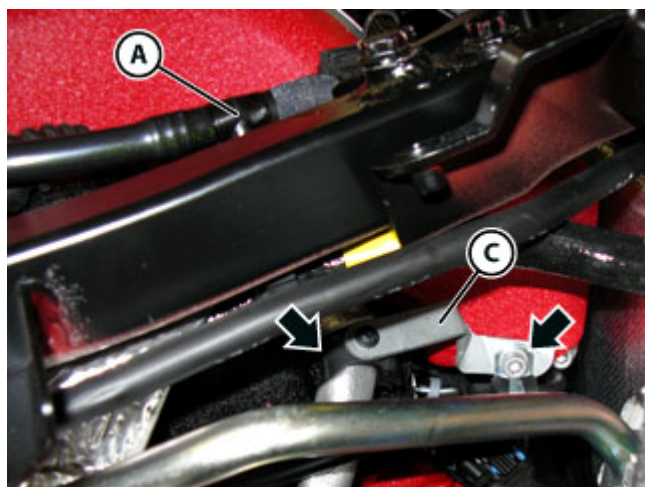
- Disconnect the pipe (A), removing the indicated collar.



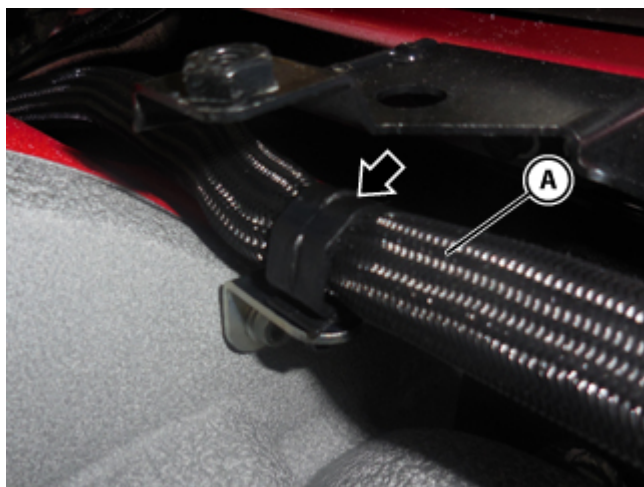
- Disconnect the pipe **(A)**, removing the indicated collar.



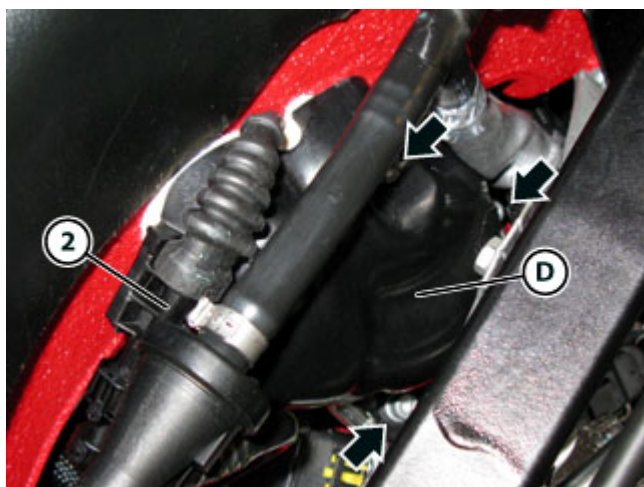
- Only for vehicles with HELE: disconnect the connector **(1)**.
- Disconnect the pipe **(B)** from the brake servo.



- Detach the pipe **(A)** from the indicated clip.
- Remove the bracket **(C)**, undoing the indicated screw.



- Detach the pipe **(A)** from the indicated clip and remove.



- Detach the connector **(2)** from the respective mounting bracket.
- Undo the screws indicated.
- Remove the soundproofing **(D)**.



- Detach the connector **(2)** from the respective mounting bracket.
- Undo the screws indicated.
- Remove the soundproofing **(D)**.



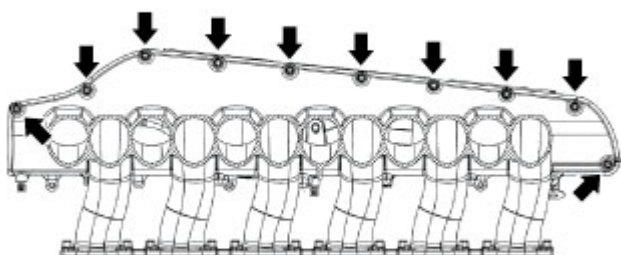
- Disconnect the pipe (E), removing the indicated collar.



- Disconnect the pipe (E), removing the indicated collar.

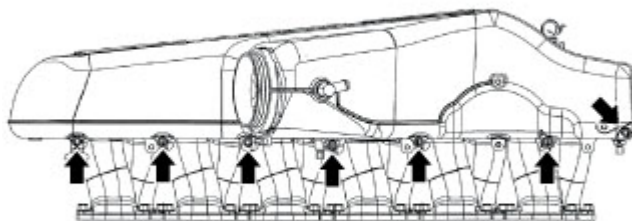
Left hand plenum, inner side

- Undo the screws indicated.



Left hand plenum, outer side

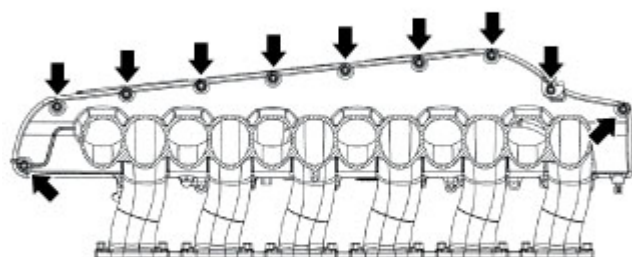
- Undo the screws indicated.



- Use adhesive tape to cover the intake manifold orifices to prevent foreign bodies from entering the engine.

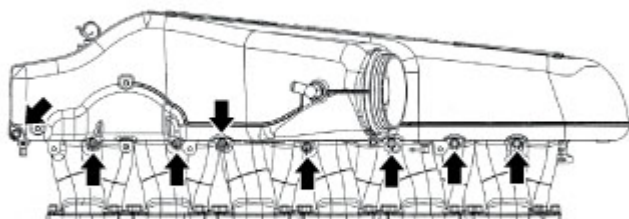
Right hand plenum, inner side

- Undo the screws indicated.



Right hand plenum, outer side

- Undo the screws indicated.



- Use adhesive tape to cover the intake manifold orifices to prevent foreign bodies from entering the engine.



- Undo the indicated screws and move the injection cable aside so that it does not hinder the subsequent operations.



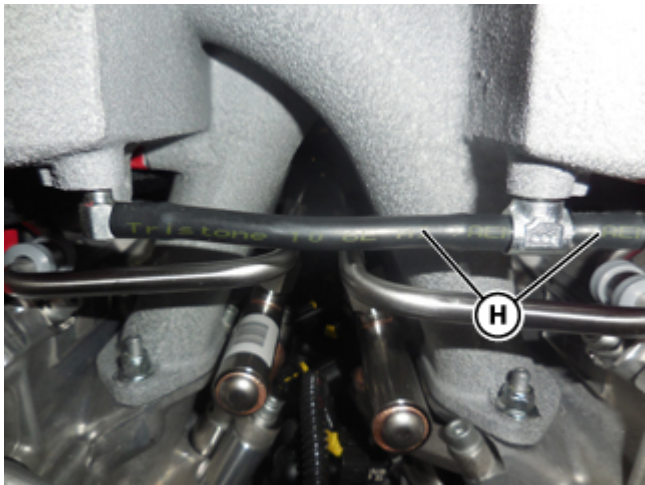
- Undo the indicated screws and move the injection cable aside so that it does not hinder the subsequent operations.



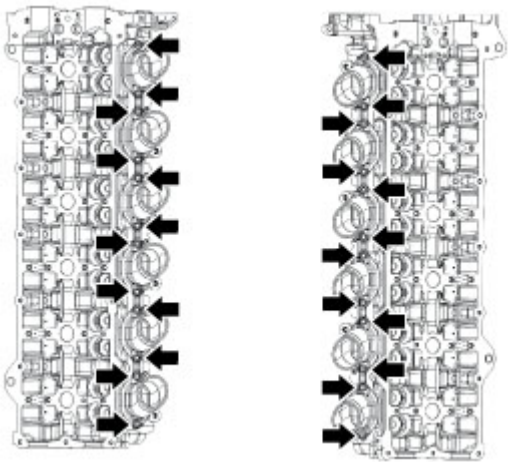
- Disconnect the pipe (G) undoing the indicated union.



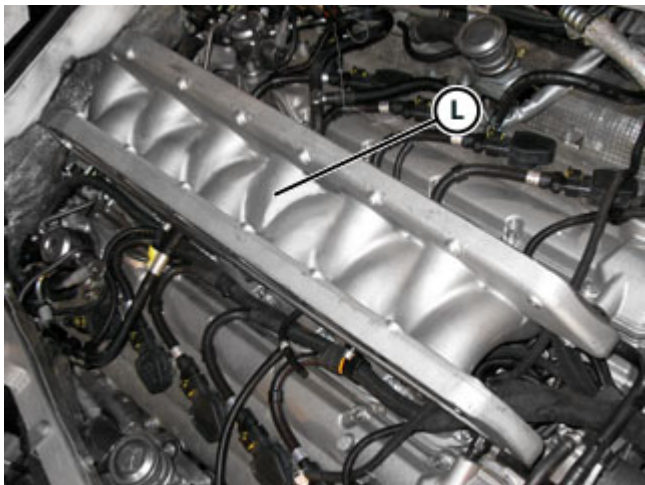
- Pull out the pipe (G) and move it aside so that it does not hinder the subsequent operations.



- Disconnect the pipes (H).



- Undo the nuts indicated.



Before removing the intake manifold, carefully check that all the nuts and washers removed have been retrieved so that they cannot accidentally enter the engine.

- Remove the intake manifold (L).

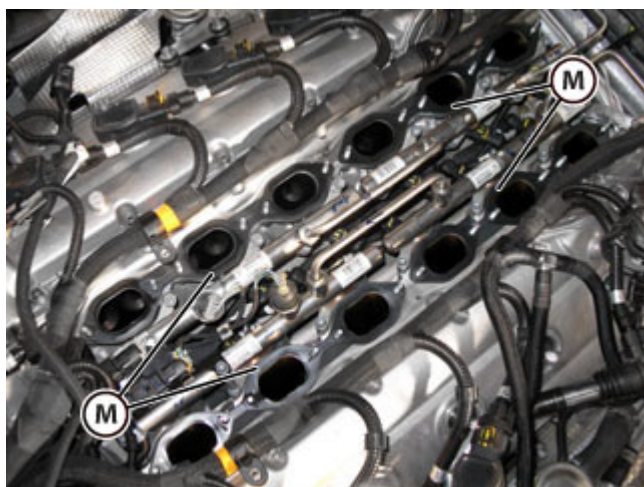
- Cover the engine intake duct orifices to prevent foreign objects from entering the engine.

Refitting the intake manifold

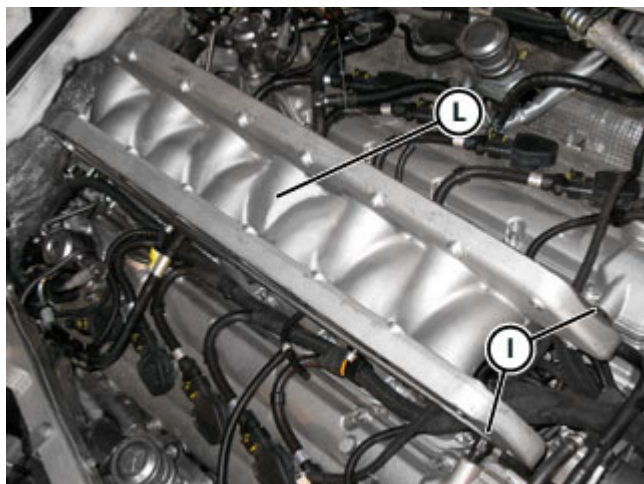
			
Tightening torque		Nm	Class
Fastening intake manifold	Nut	15 Nm	B

			
Tightening torque		Nm	Class
Fastening low pressure fuel pipe to high pressure fuel pumps	Union	20 +2 Nm -0	0
Fastening intake plenum	Screw	10 Nm	B
	Screw	12 Nm	B
Fastening soundproofing covers for GDI pumps	Nut	10 Nm	B
	Screw	10 Nm	B

- Remove the protective covering from the intake duct orifices on the engine, taking care not to allow foreign objects into the engine itself.

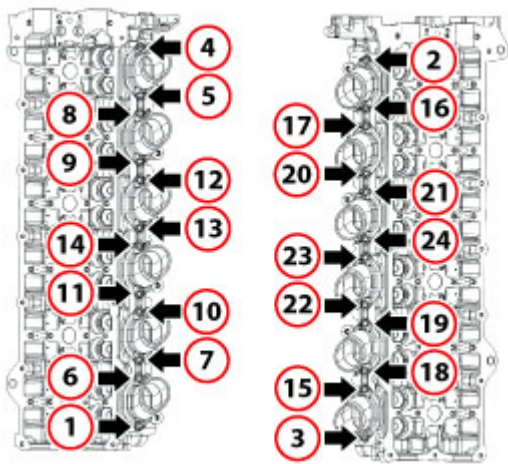


- Thoroughly clean the mating surfaces between the intake manifold and the cylinder heads.
- Replace and refit the gaskets (**M**).



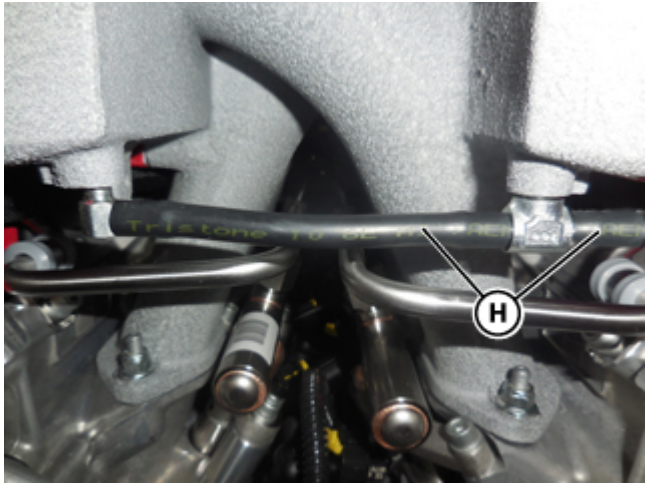
-  Fitting the intake manifold incorrectly may deform and/or damage the respective gaskets and compromise seal integrity. Check carefully that the intake manifold is fitted in the correct position.

- Fit the intake manifold (**L**).
- Replace and refit the gaskets (**I**).



- Tighten the nuts indicated.
 - Tighten the indicated nuts in the sequence given.
- i* The image shows the cylinder banks seen from above, with the front of the vehicle at the top.

		
Tightening torque	Nm	Class
Nut	15 Nm	B



- Connect the pipes (**H**).



- Fit and insert the pipe (**G**).



- Connect the pipe (G), tightening the indicated union.

		
Tightening torque	Nm	Class
Union	20 +2 Nm -0	0



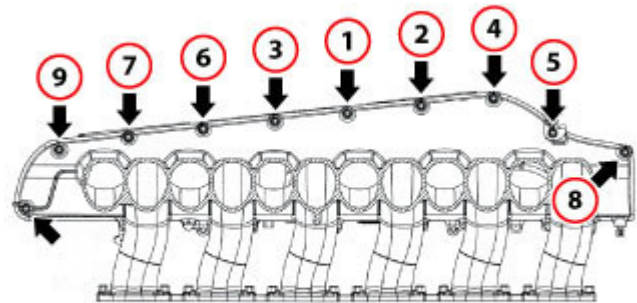
- Fit the injection cable and hand-tighten the indicated screws.



- Fit the injection cable and hand-tighten the indicated screws.



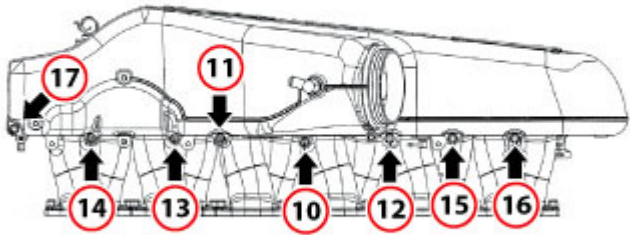
- Remove the adhesive tape from the orifices on the intake manifold, taking care not to allow foreign objects into the engine itself.



Right hand plenum, inner side

- Apply a suitable quantity of **AREXONS SYSTEM 52A43 - MEDIUM STRENGTH THREAD LOCK** to the threads of the indicated screws.
 - Tighten the screws indicated.
 - Tighten the indicated screws in the sequence given.
- i* The sequence indicated continues in the following image.

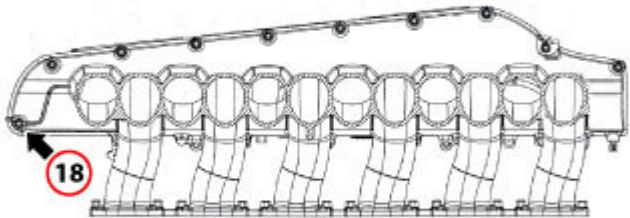
		
Tightening torque	Nm	Class
Screw	10 Nm	B



Right hand plenum, outer side

- Apply a suitable quantity of **AREXONS SYSTEM 52A43 - MEDIUM STRENGTH THREAD LOCK** to the threads of the indicated screws.
 - Tighten the screws indicated.
 - Tighten the indicated screws in the sequence given.
-  The sequence indicated continues in the following image.

		
Tightening torque	Nm	Class
Screw	12 Nm	B



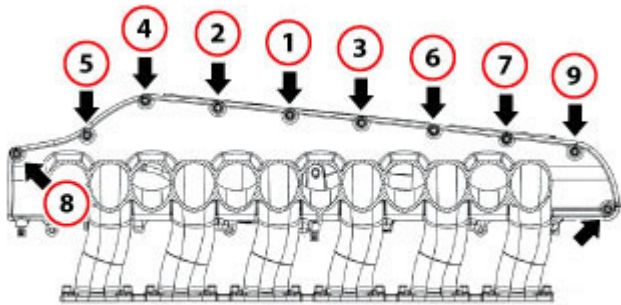
Right hand plenum, inner side

- Apply a suitable quantity of **AREXONS SYSTEM 52A43 - MEDIUM STRENGTH THREAD LOCK** to the thread of the indicated screw.
- Tighten the indicated screw.

		
Tightening torque	Nm	Class
Screw	10 Nm	B



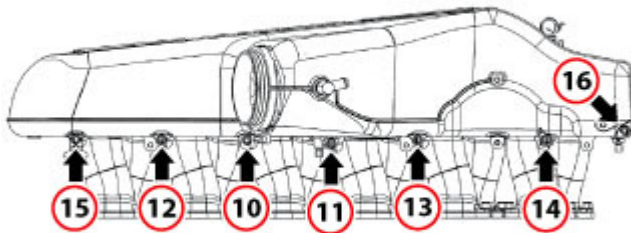
- Remove the adhesive tape from the orifices on the intake manifold, taking care not to allow foreign objects into the engine itself.



Left hand plenum, inner side

- Apply a suitable quantity of **AREXONS SYSTEM 52A43 - MEDIUM STRENGTH THREAD LOCK** to the threads of the indicated screws.
 - Tighten the screws indicated.
 - Tighten the indicated screws in the sequence given.
- i* The sequence indicated continues in the following image.

		
Tightening torque	Nm	Class
Screw	10 Nm	B

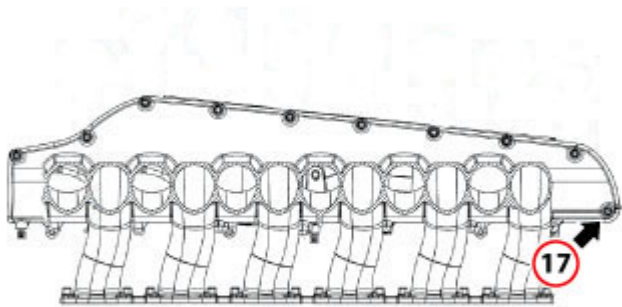


Left hand plenum, outer side

- Apply a suitable quantity of **AREXONS SYSTEM 52A43 - MEDIUM STRENGTH THREAD LOCK** to the threads of the indicated screws.
 - Tighten the screws indicated.
 - Tighten the indicated screws in the sequence given.
- i* The sequence indicated continues in the following image.

		
Tightening torque	Nm	Class
Screw	12 Nm	B

Left hand plenum, inner side



- Apply a suitable quantity of **AREXONS SYSTEM 52A43 - MEDIUM STRENGTH THREAD LOCK** to the thread of the indicated screw.
- Tighten the indicated screw.

Tightening torque	Nm	Class
Screw	10 Nm	B



- Connect the pipe **(E)**, fitting and tightening the collar as indicated.



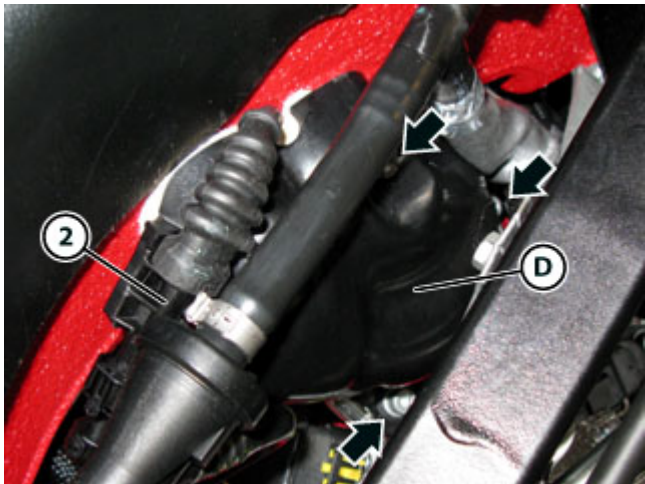
- Connect the pipe **(E)**, fitting and tightening the collar as indicated.



- Fit the soundproofing (D).
- Tighten the indicated nut and screws.

		
Tightening torque	Nm	Class
Screw	10 Nm	B
Nut	10 Nm	B

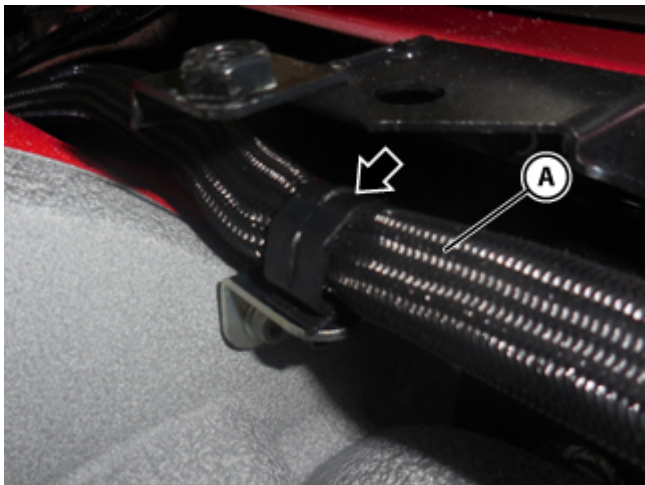
- Attach the connector (2) to the respective mounting bracket.



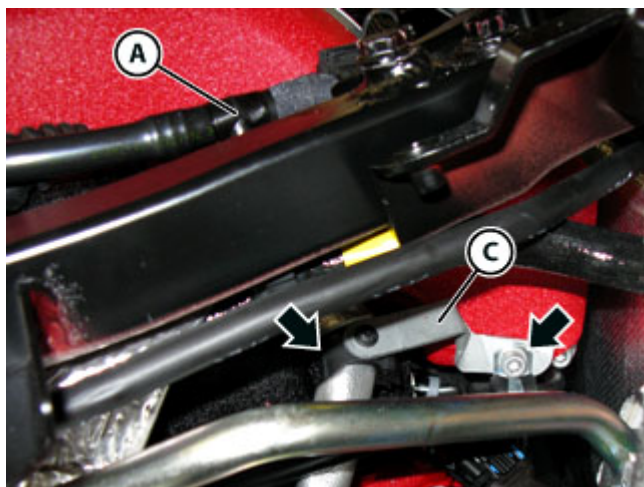
- Fit the soundproofing (D).
- Tighten the indicated nut and screws.

		
Tightening torque	Nm	Class
Screw	10 Nm	B
Nut	10 Nm	B

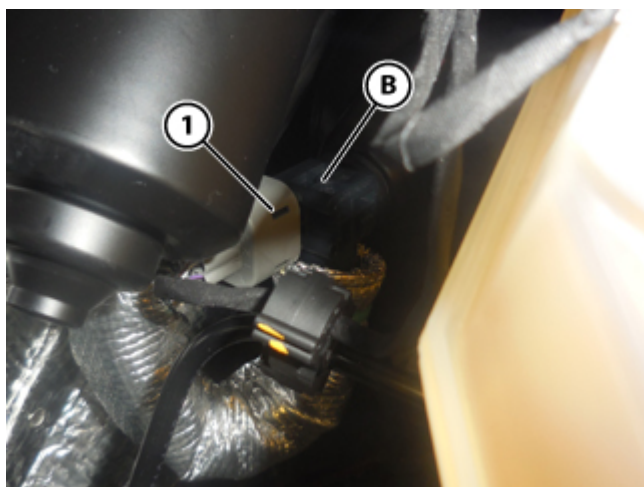
- Attach the connector (2) to the respective mounting bracket.



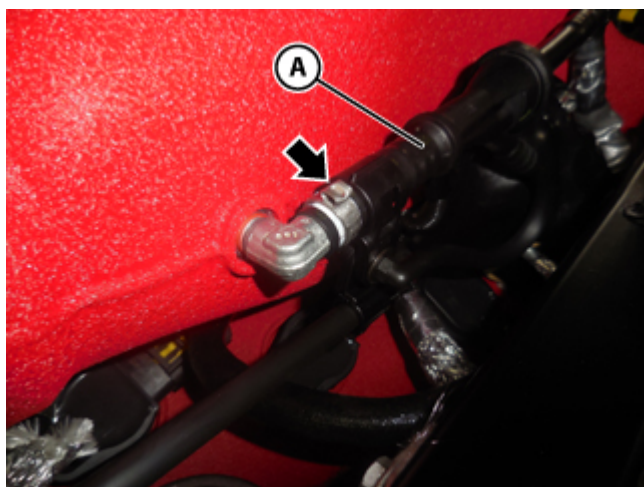
- Fit the pipe (A), fastening to the indicated clip.



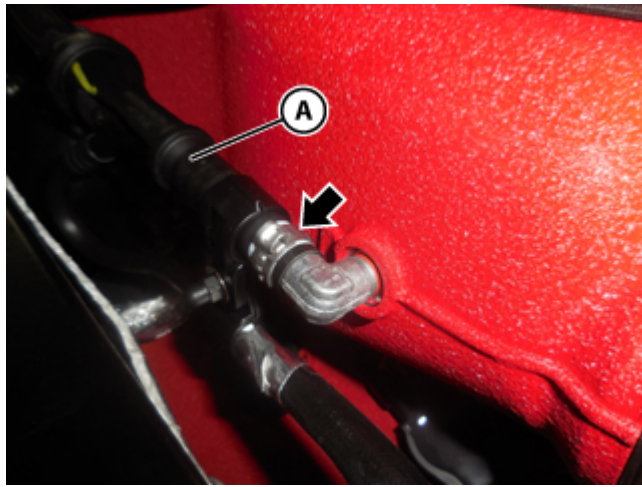
- Fit the bracket **(C)** and tighten the indicated screw.
- Fasten the pipe **(A)** to the indicated clip.



- Connect the pipe **(B)**.
- For vehicles with HELE: connect the connector **(1)**.



- Connect the pipe **(A)**, fitting and tightening a new collar as indicated.



- Connect the pipe **(A)**, fitting and tightening a new collar as indicated.

- Refit the windscreen wiper unit (🔧 E3.10).
 - Refit the motorised throttle bodies (🔧 F3.09).
 - Refit the air flow meters (🔧 F3.08).
 - Refit the engine compartment cosmetic shields (🔧 E3.14).
 - Refit the lateral underwindscreen cosmetic shields.
-
- Connect the battery (🔧 F2.01).